GM UNIVERSITY
Innovating Minds

GM UNIVERSITY

P.B. Road, Davangere - 577006

STUDENT AFFAIRS
GM UNIVERSITY

IGNITRON
2K26

THE NATIONAL LEVEL TECH FEST

IGNITRON 2K26

TECHNOLOGY FOR HUMAN FLOURISHMENT

CLOUDNEX

NEXT-GEN CLOUD ENGINEERING

RULE BOOK



1. Event Description

CLOUDNEX – Next-Gen Cloud Engineering Challenge is a dynamic, multi-round technical competition designed to test participants in the core domains of modern cloud computing. The event focuses on real-world problem solving, where participants will design scalable cloud architectures, build and deploy applications, and handle unexpected system challenges under pressure.

Through a structured format of design, development, and crisis management, CLOUDNEX simulates the role of a cloud engineer in real industry scenarios.

2. Objectives

- Promote innovation in cloud computing, system design, and architecture.
- Provide hands-on experience in building, deploying, and managing cloud applications.
- Develop skills in scalability, performance optimization, security, and fault tolerance.
- Encourage practical problem-solving, teamwork, and clear technical communication.

3. Eligibility and Team Formation

- The event is open to students across India.
- Each team must consist of 3–4 members.
- Each team should nominate one member as the primary point of contact for official communication.
- Registration details must be accurate. Any change in team membership must be approved by the event coordinators.

- Team members may be from the same or different institutions. Each member must carry a valid college ID card on the event day.

4. Registration Details

- **Registration link:** To be announced.
- **Registration fee:** To be announced.
- **Registration deadline:** To be announced.

5. Attendance, Availability and Event Conflicts

All registered participants are expected to remain available for the complete event duration and actively participate in their assigned rounds. Teams should plan their schedules in advance and avoid registering for another event or activity that overlaps with CLOUDNEX 2K26.

- If a participant has registered for another event, competition, or activity that may conflict with CLOUDNEX, they must inform the Faculty Coordinator or a Student Coordinator in advance—preferably before the event day.
- Teams must notify coordinators as early as possible if any member anticipates a scheduling conflict, late arrival, or inability to participate. This helps organizers plan rounds and avoid disruption.
- Failure to communicate a conflict in advance may affect the team's participation, timing, or eligibility at the organizers' discretion.

6. Event Duration and Format

The event is a one-day event of approximately 6 hours. Problem statements and round-specific instructions will be released according to the event schedule. The indicative time allocation is:

- **Round 1 – Architecture Design:** 45 minutes
- **Round 2 – Build & Deploy:** 3 hours 45 minutes
- **Evaluation and elimination:** 30 minutes
- **Round 3 – Cloud Crisis Challenge:**1 hour30 minutes(qualified teams only)

Exact timings and reporting time will be shared by the organizers before the event.

The competition consists of three progressive rounds:

Round 1 — Architecture Design: Polesitter Circuit Blueprint

Teams design a secure, scalable, fault-tolerant, and cost-efficient cloud architecture for the assigned problem. The design should explain proposed services, network boundaries, data flow, scalability, disaster recovery, and estimated costs.

Expected output: An animated GIF architecture diagram that shows the request flow step by step (user → edge/CDN → API → services → database/storage → monitoring), labelled with the chosen cloud provider's services, the auto-scaling approach and the failure/backup path. Teams also submit a one-page scope sheet covering technology choices, security and scalability strategy, and a cost estimate.

- Diagrams may be made in draw.io, Excalidraw, Lucidchart or a similar tool and exported as a GIF, or recorded as a screen capture and converted to GIF.
- The GIF should not exceed 2 MB and must be submitted through the method announced by the organizers.

Round 2 — Build & Deploy: Pit Stop Deployment

Teams build and deploy the same problem statement and architecture they submitted in Round 1. Any change from the Round 1 design must be explained and justified during the demonstration. Teams must demonstrate the working solution and explain their deployment approach.

Expected output: A live application on a public HTTPS URL hosted on AWS, Azure or GCP; a public GitHub repository; a README (problem, architecture GIF, tech stack, demo login details and setup steps); and a short technical demonstration. Demos running only on localhost will not be accepted.

Round 3 — Cloud Crisis Challenge: The Final Lap

Qualified teams respond to a controlled cloud incident or simulated failure, explain how their solution maintains reliability and availability. Crisis scenarios will be conducted in approved test environments.

Expected output: Demonstration of troubleshooting or recovery, the mitigation applied, and a short explanation of the solution.

Problem Statements

Problem statements are scenario-based and grouped into three levels: Easy, Intermediate and Hard (15 problems each). All problems within a level carry equal weightage and are evaluated against the same judging criteria.

- **Selection:** Teams choose one problem statement at the start of Round 1, using the selection method announced by the organizers. The same problem statement is carried through all rounds.

- **Limit:** The organizers may limit how many teams can take the same problem statement.
- **Changes:** Problem statements cannot be changed once Round 1 begins.

7. Elimination Rule

After Round 2, teams will be evaluated and an elimination decision will be made. Round 1 and Round 2 scores are combined for this decision. Only the top-performing teams selected by the judges will advance to Round 3. The number of qualifying teams will be announced by the organizers.

8. Judging Criteria

- **Innovation & Design:** Creativity, suitability, and quality of the architecture or solution.
- **Functionality:** Working implementation and successful demonstration/deployment.
- **Scalability & Optimization:** Efficient resource use and ability to handle increased demand.
- **Security & Reliability:** Appropriate protection, fault tolerance, and recovery planning.
- **Problem-Solving:** Troubleshooting, decision-making, and response to the assigned challenge.
- **Presentation & Clarity:** Clear explanation, teamwork, and justification of technical decisions.
- **Tie-breaker:** If teams are tied, the team with the higher Round 3 score ranks higher; if still tied, the higher Functionality score decides.

The judges' decision is final and binding.

9. Optional Bonus

Relevant and functional blockchain integration may earn bonus points at the judges' discretion. Teams must explain why blockchain is appropriate for their solution; merely mentioning or adding an unrelated blockchain component will not guarantee bonus points.

10. Cloud Accounts and Technical Preparation

- Participants should create and test any required cloud-provider accounts before the event.
- Teams may use any one of Amazon Web Services (AWS), Google Cloud Platform (GCP) or Microsoft Azure as their primary cloud provider. Using services from more than one provider is allowed but must be shown clearly in the architecture.
- Participants are responsible for account access, credentials, quotas, and any cloud usage charges. Organizers cannot guarantee that a third-party provider will approve or activate an account during the event.
- Free-tier or student credits are recommended. Teams should set a billing/budget alert before the event and delete or stop all resources after the event to avoid charges.
- Teams must bring their own laptops and chargers. A mobile hotspot is recommended as a backup internet connection.

11. Submission and Team Conduct

- Submit work using the method and before the deadline announced by the organizers.
- Ensure submitted links and repositories are accessible to judges and do not expose secrets or private personal information.

- Use only dummy or synthetic data. Real personal, health, financial or identity data must not be used.
- AI assistants (such as ChatGPT, Claude or GitHub Copilot) may be used. Teams must be able to explain every part of their solution, and any AI-generated feature in the application must be labelled as AI-assisted.
- Open-source libraries and starter templates are allowed, but the solution itself must be built during the event. Projects prepared before the event will be disqualified.
- Teams must present their own work, acknowledge third-party libraries/resources, and avoid plagiarism, unauthorized access, or attempts to interfere with another team's systems.
- Testing or failure simulation must remain within the team's own application or an environment explicitly approved by the organizers. Do not attack public services or other participants' infrastructure.
- Follow instructions from faculty, student coordinators, volunteers, and judges. Organizers may address rule violations or disruptive conduct, including disqualification in serious cases.

12. Event Date and Venue

Date: To be announced / confirmed by the organizers.

Reporting time: To be announced by the organizers.

Venue: GM Business School Auditorium, New Building, GM University, Davangere, Karnataka.

13. Prizes and Certificates

- Exciting prizes will be awarded to the winning teams, as announced by the organizers.
- Participation certificates will be provided in accordance with the event's completion and attendance requirements.

14. Event Coordinators

Student Coordinators:

- Mr. Vinayak Pujar – 8088079573
- Mr. Darshan G M – 7676566133

Faculty Coordinator:

- Ms. Usha T M – 6361736961
- Ms. Pooja K R – 6363725690

Institution: GM University, P.B. Road, Davangere, Karnataka.

The organizers may clarify or update operational details when necessary. Important changes will be communicated through official event channels.